

420-D03-AB — CYBER DEFENSE

LAB 01 — Introduction to Cyber Defense: Defense Methods and Zoning Concepts, with AI Tools

Field	Value
Programme	Cybersecurity LEA.D8 — AEC (Attestation d'études collégiales)
Course / Ponderation	420-D03-AB Cyber Defense — 2-3-3 — 3.00 credits
Competency	EZ03 — apply access control measures. This lab: elements 1.1, 1.2, 1.3
Sessions	S01 Sat 29 Aug 2026 (Intro, Defense Methods) · S02 Tue 1 Sept 2026 (Zoning, Q_01)
Weight	5 % — one of six labs ($6 \times 5 \% = \mathbf{30 \%}$, Continuous Course Assessments)
Effort	3 hours minimum of hands-on work, plus the write-up
DUE	Sunday 13 September 2026, 23:59 (midnight) EDT — Moodle → <i>Lab 01 DropBox</i>
Late penalty	–10 % per calendar day (department policy)
Work mode	Individual. Discuss approaches freely; every artefact is your own.
Prepared by	Cyber.SoHo Educational Hub — v1.0, August 2026

§0 — STOP. READ THIS BEFORE YOU TOUCH A KEYBOARD.

Some commands and scripts here are close cousins of the ones a real attacker uses. The difference between a technician and a criminal is **not skill**. It is **authorisation**.

0.1 — What you MAY do

- Run every supplied script **on a machine you personally own**, or on **your own isolated pod** on the CyberQuébec range.
- Inspect, break, rebuild and delete **your own** virtual machines.
- Replace any tool with a different one (see §7), provided you write down what and why.
- Ask a **locally installed** AI (Artificial Intelligence) model for help, under the rule in §0.4.

0.2 — What you MUST NOT do ✗ — no exceptions, no "just to see"

Forbidden	Why
Scan, probe, log into or "test" any machine that is not yours — a classmate's pod, the range's management plane, the College network, your workplace, a friend's shop, your landlord's Wi-Fi	Criminal offence (§0.3). Also an automatic zero and a referral under College policy.
Run these scripts on an employer's equipment without a signed, dated, scoped authorisation	"My boss said it was fine" is not a scope document. Get it in writing, or do not run it.
Upload logs, configurations, screenshots, IP (Internet Protocol) addresses or credentials to any hosted AI service	A security log is a reconnaissance report about its own owner. Local models only.
"Hack back", retaliate, or investigate a machine that appears to be attacking you	Criminal in Canada regardless of provocation. Stop, document, report to a named person in writing .
Put any real password, private key or personal information in your archive	Name the secret and where it lives. Never the value.
Change the range's shared infrastructure, quotas or another student's resources	Not yours. Not in scope.

The one-sentence rule for your whole career: *your authority ends at your own boundary, and your boundary is defined by a contract — not by a router.*

0.3 — The law you are working under

Unauthorised access to, interference with, or interception of a computer system or its data is a **criminal offence**. Read at least the first link before the next class.

- **CA Criminal Code of Canada**, s. 342.1 (unauthorised use of a computer) and s. 430(1.1) (mischief in relation to computer data) — <https://laws-lois.justice.gc.ca/eng/acts/C-46/section-342.1.html>
- **US United States** — Computer Fraud and Abuse Act (CFAA), 18 U.S.C. §1030 — <https://www.law.cornell.edu/uscode/text/18/1030>
- **EU European Union** — Directive 2013/40/EU on attacks against information systems — <https://eur-lex.europa.eu/eli/dir/2013/40>
- **ES Spain** — Ley Orgánica 10/1995, del Código Penal — <https://www.boe.es/buscar/act.php?id=BOE-A-1995-25444>
- **CA Personal information** — Quebec Law 25 — <https://www.cai.gouv.qc.ca> · federal PIPEDA (Personal Information Protection and Electronic Documents Act) — <https://www.priv.gc.ca>

-  Free national guidance for the size of business you will defend — **Canadian Centre for Cyber Security** — <https://www.cyber.gc.ca>

Ignorance of scope is not a defence. If you are not certain an action is authorised, then it is not. Stop and ask in writing.

0.4 — The AI rule for this course

The model drafts. The analyst decides. The analyst's name goes on the record.

1. **Local only.** Your model runs on your own machine, bound to 127.0.0.1 — the *loopback* address, where traffic physically cannot leave the computer. Recommended: **Ollama** (<https://ollama.com>) with a small model such as llama3.2:3b.
2. **Redact before you prompt.** Strip e-mail addresses, phone numbers, passwords, tokens and hashes *before* any text reaches the model. Prompts get logged.
3. **Declare it.** One line per submission: which tool, which model, and **what you changed after checking**. Undeclared AI text is an academic-integrity matter under **John Abbott College Policy No. 7 (IPESA)** — <https://www.johnabbott.qc.ca>.
4. **No model, no problem.** Every task below has an offline path. A laptop with 8 GB of RAM (Random Access Memory) and no graphics card completes this lab in full, for full marks.

0.5 — Acknowledgement — a graded deliverable

Copy this into LAB_01/00_ROE_ACKNOWLEDGEMENT.md, complete it, include it in your archive. **A submission without it is marked incomplete.**

RULES OF ENGAGEMENT — ACKNOWLEDGEMENT

Course: 420-D03-AB Cyber Defense Lab: 01 Cohort: CYB-01

Name: _____ Student number: _____ Date: _____

I confirm I have read Section 0 of the Lab 01 brief in full.

Every command in this submission was executed ONLY on:

☐ my own personal equipment, and/or

☐ my own isolated pod on the CyberQuebec range.

I did not access, scan or test any system belonging to another person, student, employer or institution.

No real credential, private key or personal information appears in this archive.

AI declaration: tool _____ model _____

What I changed after verifying the output: _____

Signature (typed full name): _____

§1 — THE LAURENTIDE THREAD — why a 5 % lab is really a 55 % lab

Read this twice. It is the one thing students discover too late.

Your six labs, your Final Project and your Course Procedure Manual are **not eight assignments**. They are **one estate, built once and graded three times**. That continuous body of work is **the Laurentide Thread**, named after your client.

Your client, today until 24 October: *Laurentide Outfitters inc.*, a fictional Montréal outdoor-equipment retailer. Thirty-five staff, one head office with a small warehouse, a public online shop, customer records under Quebec Law 25. Their network grew the way small networks grow: one device at a time, no plan, everything trusting everything. Their brief: *"Make our network defensible. Prove it works. And leave us a manual so the next person doesn't have to call you."*

Pass	What it builds	%	Lands in
LAB 01 (this one)	Zones, addressing plan, traffic matrix, entry-point register	5	R1 — the skeleton everything hangs on
LAB 02	Identity plane — accounts, groups, least privilege	5	R2
LAB 03	Host hardening, host firewalls, before/after scores	5	R2
LAB 04	Detection — IDS/IPS (Intrusion Detection / Prevention System), agents, triage	5	R3
LAB 05	Network device security, secure protocols, management plane	5	R3, R6
LAB 06	Cryptography — CA (Certificate Authority), TLS, VPN, encryption at rest	5	R4
FINAL PROJECT	The six labs assembled into one estate and re-tested end to end	15	Wed 21 Oct 2026
PROCEDURE MANUAL	The same work again, rewritten as procedures a stranger can follow	10	Wed 21 Oct 2026

Total riding on one estate: 55 %. Three consequences to act on tonight:

1. **The Final Project is not a new build.** It is these six labs consolidated. If your labs contradict each other — and they will, if built carelessly — October is where that surfaces, with no time left.
2. **The Manual is not a description.** *A procedure has a pre-condition, numbered steps, a verification test, and a rollback. If you cannot say how to undo it, you have written a diary entry.*

3. **Write the lab up the same evening you do it.** Forty minutes per lab and you finish the term calmly. Skip it, and you spend mid-October trying to remember what a 12 September screenshot proved.

The classmate test decides every sentence from now on: *could a classmate who was not with you rebuild and verify your estate from your Manual, without asking you one question?*

§2 — WORDS YOU NEED FIRST (plain-language glossary)

No IT (Information Technology) background is needed to pass this lab. These fifteen words are. Every abbreviation in this course is expanded in round brackets on first use — do the same in your own writing. It is a house rule and it is graded.

Term	In plain words
Host	Any single computer on a network — laptop, server, printer, phone.
VM (Virtual Machine)	A whole computer that exists as a file on a real computer. Safe to break and delete.
IP address / subnet (Internet Protocol)	A machine's postal address, e.g. 172.16.30.15. A subnet is a street of them: 172.16.30.0/24 = the 254 usable addresses .1 to .254.
Port / service	A port is a numbered door on one host (443 = secure web, 445 = file sharing, 22 = remote login). A service is the program behind that door, waiting for a knock.
Listening	A service with its door open. Something <i>listening</i> is something reachable.
0.0.0.0	Not an address — a wildcard meaning " every network card on this machine. " A service on 0.0.0.0 is reachable from the guest Wi-Fi.
127.0.0.1 (loopback)	"This machine only." Traffic here never touches a network card.
Firewall	A guard at a doorway, allowing or blocking traffic by written rules.
VLAN (Virtual Local Area Network)	A way to split one physical switch into several separate networks.
Zone	Systems sharing the same trust level and access rules , separated from other zones by a control point that inspects traffic between them . The control point is the important half.
Default deny	The last rule blocks everything. Every allowed flow is therefore a decision you can name.
Attack surface / entry point	The surface is the geography — <i>all</i> the ways in. An entry point is one specific door.
Evidence	A file that proves a claim. "Firewall screenshot" proves nothing; " <i>WAN blocks inbound TCP 3389 after rule 4</i> " does.

⚠ **Two VLANs with a router between them and an "allow any" rule are not two zones.** They are one zone with extra paperwork.

§3 — SETUP (≈ 20 min)

Where to work. Either your own isolated pod on the **CyberQuébec** range — <https://documentation.cyberquebec.org>, shelf *Guide de l'enseignant*, book *Gestion de laboratoire* — or a free VM on your own laptop (VirtualBox, KVM, Proxmox VE). **Both are acceptable for Lab 01.**

Before opening any link or downloading any file in this course, check it with a reputation service such as <https://www.virustotal.com/gui/home/url>. Not a formality — it is the criterion "*consideration of relevant sources of information*" applied to your own browsing.

3.1 Build your workspace. Download `rle01_cpm_scaffold.py` from the Moodle Class 01 folder and run:

```
python3 rle01_cpm_scaffold.py --student "Firstname Lastname"
```

This creates `CPM_Firstname_Lastname/` with a cover page, an `EVIDENCE_REGISTER.csv`, and one folder per lab carrying the seven mandatory Manual sections. **That folder is your Course Procedure Manual.** Keep it in **one** location, under version control if you can — never in two synced cloud folders edited from two machines.

3.2 Open `00_COVER.md`; add your student number, cohort `CYB-01`, and today's date.

3.3 Optional (15 min). Install Ollama (<https://ollama.com>), then `ollama pull llama3.2:3b`.

§4 — THE WORK

Evidence naming standard — from the first screenshot.

`CYB01_LASTNAME_LAB01_EVIDyy_YYYYMMDD_short-description.ext` — e.g. `CYB01_TREMBLAY_LAB01_EVID03_20260906_zone-matrix.png`. Every screenshot shows hostname, date/time and your own session, undoctored. Log each item in `EVIDENCE_REGISTER.csv` as you go; aim for **4–6 strong items**. Curated evidence proves judgement; screenshot spam proves you found the screenshot key.

TASK 1 — Count the doors before you buy locks (≈ 35 min)

EZ03 1.1 — accurate identification of perimeter elements.

You cannot defend a door you never counted. Before any zoning or firewall: **produce a written register of what your own machine currently offers to the network.**

1.1 Download `rle03_entry_point_inventory.sh` and run it on your lab VM:

```
bash rle03_entry_point_inventory.sh --out
LAB_01/evidence/entry_points_before.csv
```

1.2 Read the output, do not skim it. Every row is one open door. The `exposure` column is what matters:

Value	Meaning	What you do
LOOPBACK_ONLY	Reachable only by this machine	Nothing. Lowest exposure.
SPECIFIC_INTERFACE	Reachable on one network card	Confirm that card faces the zone you intended.
ALL_INTERFACES	Bound to 0.0.0.0 — reachable from everywhere, guest Wi-Fi included	Justify it in writing, or bind it to 127.0.0.1.

1.3 Make a door appear, on purpose. In a second terminal start a throwaway web server, then re-run:

```
python3 -m http.server 8080          # leave running
bash rle03_entry_point_inventory.sh --out
LAB_01/evidence/entry_points_after.csv
```

A new `ALL_INTERFACES` row appears on port 8080. **Screenshot both files side by side.** You just watched a routine action of your own become a documented entry point in ten seconds. `Ctrl+C` to stop.

1.4 Write the register. For every `ALL_INTERFACES` row, one sentence answering exactly this: "Who needs to reach this service, and from which zone?" If you cannot answer, the correct entry is *"No identified consumer — candidate for shutdown or binding to loopback."* That is a **finding**, and findings earn marks.

1.5 Now do Lauretide Outfitters — on paper, no commands. From §1, list **at least eight** entry points, each classified *external path* or *internal device*. A fourteen-person accounting firm had eight; a thirty-five-person retailer with an online shop has more. Think: the printer, the boardroom television, the heating controller the landlord installed, the bookkeeper's connection, staff phones.



Produces: `entry_points_before.csv`, `entry_points_after.csv`,
`ENTRY_POINT_REGISTER.md`, 2 screenshots.

TASK 2 — DEFENSE METHODS, WITH AI TOOLS (≈ 40 min)

A raw CSV (Comma-Separated Values) file is not a report. A client cannot read it, and neither can you at two in the morning. A local model produces a **first draft** — then you do the part it cannot: **verify it**.

2.1 Classify by hand first — 10 min, no computer. Take any **five** services from your Task 1 register and fill one row each. You must be able to do this unaided, because in 2.3 you will be checking a machine that cannot.

Service / port	Which of the 7 layers?	Function: <i>deterrent · preventive · detective · corrective · compensating</i>	Type: <i>administrative · technical · physical</i>	If removed, what breaks?
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2.2 Redact, then draft. Run the triage assistant against the supplied sample alert:

```
python3 rle02_ai_triage_assistant.py --alert sample_alert.json #
with a local model
python3 rle02_ai_triage_assistant.py --alert sample_alert.json --offline #
no model needed
```

Open the decision record it writes. Note three things, one sentence each, in your findings: the model's output sits under a key literally named **draft_not_verdict** (naming is a control); four analyst fields are **deliberately empty**, so the record is structurally incomplete until a person types their name into it (accountability as a data structure); and redaction happens **before** anything reaches the model — confirm it with `grep -i "password\|@" sample_alert.json`, then compare against the sanitised prompt.

2.3 The actual skill — catch the model being wrong. Run `--offline` **first**, then with the model, and compare. The offline draft is duller, shorter, and **never wrong**. The model draft reads better and, about one time in ten, asserts something the alert does not support: an invented hostname, a time not in the data, a confident tactic name with nothing behind it. Finding that one time is the skill. **Deliverable — a three-column table. Do not skip a row because it looks right.**

Claim the model made	In the raw data? (yes / no / partly)	My correction, or "confirmed — field X, line Y"
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2.4 Feed it your own work. Paste your **redacted** Task 1 register into the model, ask for a plain-English summary "for a non-technical business owner", then verify every sentence against your CSV as above. No model? Write the summary yourself and say so in your declaration — you lose nothing.

 That register names your hosts, addresses and open services. It is a reconnaissance report about your own machine. It goes to 127.0.0.1 and nowhere else. **Ever.**

 **Produces:** DEFENSE_METHODS_CLASSIFICATION.md, the decision-record JSON (JavaScript Object Notation) file, AI_VERIFICATION_TABLE.md, your AI declaration line.

TASK 3 — ZONING: TURN A FLAT NETWORK INTO A DEFENSIBLE ONE (≈ 45 min)

EZ03 1.2, 1.3 — the heart of the lab and the skeleton of your Final Project.

Everything so far was counting. Now you design. **No configuration yet** — Lab 05 builds it. Today you produce the document Lab 05 implements. That order is deliberate: professionals design before they type.

3.1 The zone table. Build it for Laurentide Outfitters. Addressing below is the course reference plan; you may adjust within **RFC 1918** private ranges (<https://www.rfc-editor.org/rfc/rfc1918>) if you write down why.

Zone	Reference subnet	What lives here	Trust	Internet?
WAN	203.0.113.0/24 (simulated, RFC 5737)	The outside world	None	—
DMZ (Demilitarised Zone)	172.16.10.0/24	Public web shop (app-01)	Low — assume hostile	Inbound
SRV (Servers)	172.16.20.0/24	Directory, DNS (Domain Name System), files (dc-01)	High	Outbound only
LAN (Local Area Network)	172.16.30.0/24	Staff workstations (ws-01)	Medium	Outbound
MGMT (Management)	172.16.99.0/24	Monitoring, administration (mon-01)	Highest	No
UNMANAGED (you add this row)	<i>your choice</i>	Printer, boardroom TV, heating controller, personal phones	Lowest	Vendor cloud only

Why the UNMANAGED row earns marks. Those devices almost never need to reach anything inside, and frequently need a vendor's cloud service. Their own zone costs one VLAN and removes an entire category of lateral movement. The 2026 Verizon DBIR (Data Breach Investigations Report — <https://www.verizon.com/business/resources/reports/dbir/>) found **48 %** of breaches involved a third party. That heating controller *is* that statistic, made physical.

3.2 The traffic matrix. A grid: source zones down the left, destinations across the top. Fill **every** cell with `ALLOW + protocol + a one-line business reason`, or `DENY`. The bottom-right of every ruleset is the same rule: **deny everything, and log it**. Three rules you are graded on:

1. **Default deny.** The last rule denies all. Every `ALLOW` above it is an exception you can name.

2. **Name the flow, not the address.** Users → Servers, TCP 445, file sharing survives an IP change. 10.64.20.0/24 → 10.64.30.10 allow does not, and nobody in two years will know why it exists.
3. **Log the denies you care about.** Logging every dropped packet fills the disk. Logging denied traffic **between internal zones** tells you something went wrong *inside* — exactly the signal you want.

3.3 Justify five flows. Pick the five ALLOW cells you are least sure about. Two sentences each: **who needs it**, and **what breaks in the business without it**. If the second sentence is "nothing", change the cell to DENY. Not a trick — the job.

3.4 Draw it. Zones as boxes, the firewall as the single control point between them, arrows only where the matrix says ALLOW. **Hand-drawn and photographed is perfectly acceptable** — often clearer than a tool. Prefer software? draw.io, Excalidraw, Mermaid, Visio are all fine (§7).

3.5 One short paragraph, your own words. *Guest Wi-Fi and staff Wi-Fi share one VLAN at Laurentide. Name two specific bad outcomes, and the exact change that fixes both.*

 **Produces:** ZONE_PLAN.md, TRAFFIC_MATRIX.md, zone-diagram.png, your paragraph.

TASK 4 — ZONING, WITH AI TOOLS: MAKE THE MODEL ATTACK YOUR OWN DESIGN (≈ 25 min)

In Task 2 the model **drafted**. Here it does something better: **hostile review** of work you already did. That is the professional pattern — never ask the model for the answer, ask it to find the hole in yours.

4.1 Paste your traffic matrix — redacted, real addresses replaced with <SRV_NET>, <LAN_NET> — into your local model. Adapt this prompt; the wording is yours to improve:

You are reviewing a network zoning design produced by a junior technician. Do not rewrite it. Do not compliment it.
List only: (1) any ALLOW flow whose stated business reason does not justify its breadth, (2) any pair of zones that can reach each other by a path the author did not notice, (3) any single fact which, if false, would defeat more than one zone boundary at once.
For each item, state the specific cell or rule you are referring to.
If you cannot find an issue, say "none found" rather than inventing one.

4.2 Verify every objection. Real problems and invented ones arrive in the same confident tone. Record all of them:

Objection raised	Real or invented?	Evidence I checked	What I changed (or why I rejected it)
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4.3 Answer the correlated-failure question yourself. Name **one fact** about your design which, if false, breaks more than one zone boundary at once. Candidates: one administrator credential

used everywhere; a management network reachable from the user zone; one hypervisor hosting the firewall *and* the file server *and* the backup target. **Write it down, with the compensating control you would add.**

This separates an analyst from a salesperson: *an analysis that hides its assumptions is not an analysis.*

No model installed? Use the arithmetic instead: run `rle06_resilience_model.py`, set your zoning layer's effectiveness to 0.00, record how far residual risk moves, then answer 4.3 as written. Full marks.



Produces: AI_ZONING_REVIEW.md, CORRELATED_FAILURE.md, updated TRAFFIC_MATRIX.md showing what changed.

TASK 5 — BASELINE THE CLIENT (≈ 20 min)

Open `rle05_defence_readiness_console.html` by **double-clicking it**. That is the entire installation procedure — one file, no internet, nothing uploaded anywhere.

5.1 Answer all fourteen questions **honestly for your own home network**. Not aspirationally. Unknown scores zero, and that is accurate, not pessimistic: from an attacker's point of view there is no difference between a control that does not exist and one nobody can confirm exists.

5.2 Export the CSV into `LAB_01/evidence/`. **Keep it**. You run the same console again at Class 09 and Class 17, and **the trend is part of your Final Project mark**.

5.3 Four sentences: your score, the two zero-scoring rows that cost **nothing** to fix, and what you will actually do about them this month.



Produces: day01_readiness_findings.csv, READINESS_BASELINE.md, 1 screenshot of the score.

TASK 6 — WRITE THE MANUAL ENTRY (≈ 25 min) — *where marks are lost*

Open `CPM_Firstname_Lastname/LAB_01/LAB_01.md`. The seven headings are already there. Fill **all seven**. An entry missing its **Rollback** is incomplete no matter how good the other six are.

1. **Objective** — why this control exists; the risk it addresses, in the client's terms.
2. **Scope & assumptions** — hosts and zones touched; versions; what you assumed; what it does **not** cover.
3. **Pre-conditions** — what must be true before step 1: snapshot taken, credentials in hand, change window agreed.
4. **Procedure** — numbered steps, one action each, **parameterized** (`<SRV_NET>`, not `172.16.20.0/24`). A stranger types these.

5. **Verification** — the exact test proving it works, and the expected output. *An untested control is an assumption.*
6. **Rollback** — the tested way back: snapshot revert, config restore, reverse steps. **Tested, not theoretical.**
7. **Evidence** — evidence IDs from your Register, plus problems hit and how you resolved them.

 **Produces:** completed LAB_01.md, updated EVIDENCE_REGISTER.csv, a new revision-history row in 00_COVER.md.

§5 — WHAT YOU SUBMIT

One archive: CYB01_LastName_FirstName_LAB01.zip → Moodle **Lab 01 DropBox**, by **Sunday 13 September 2026, 23:59 (midnight) EDT.**

```
CYB01_LastName_FirstName_LAB01.zip
— 00_ROE_ACKNOWLEDGEMENT.md      <- $0.5, signed. Missing = incomplete.
— LAB_01.md                        <- the seven-section Manual entry
— ENTRY_POINT_REGISTER.md         <- Task 1
— DEFENSE_METHODS_CLASSIFICATION.md <- Task 2
— AI_VERIFICATION_TABLE.md        <- Task 2
— ZONE_PLAN.md    TRAFFIC_MATRIX.md <- Task 3
— AI_ZONING_REVIEW.md    CORRELATED_FAILURE.md <- Task 4
— READINESS_BASELINE.md      <- Task 5
— EVIDENCE_REGISTER.csv
— CHANGE_LOG.md              <- only if you substituted a tool ($7)
— evidence/                  <- 4-6 curated items, named to
standard
```

Never submit real secrets. No real passwords, private keys or personal data anywhere in the archive. Name *what* the secret is and *where* it lives — never the value.

§6 — HOW THIS LAB IS MARKED (5 %)

#	Criterion	%	Full marks require
1	Rules of engagement	0.5	Acknowledgement complete and signed; AI declaration present and specific
2	Entry-point identification (1.1)	1.0	Register complete; every ALL_INTERFACES service carries its "who, from which zone" sentence; Laurentide list reasoned
3	Defense methods & AI verification	1.0	Five services classified by function and type; verification table shows you actually checked — one caught error, or a stated "all confirmed, here is how"
4	Zoning design (1.2, 1.3)	1.5	Six zones including UNMANAGED; matrix complete and ending in default-deny; five flows justified in business terms; diagram matches the matrix

#	Criterion	%	Full marks require
5	Documentation quality	1.0	All seven Manual sections including a real Rollback; parameterized steps; evidence named and registered; passes the classmate test
	Total	5.0	

*Partial credit follows production logic: a control that exists but was never proven earns half; one claimed but absent earns nothing; a **documented, justified** exception earns respect.*

§7 — EVERYTHING HERE IS OPEN. CHANGE IT.

Read this as an invitation, because it is one. Every tool, script, technique, address plan and most of the exercise content here is **open and substitutable**. Ollama, the Bash inventory script, the HTML console, the diagram tool, the operating system, the addressing scheme — **all of it can be replaced** by anything achieving the same outcome that you can get hold of.

Better still: write your own. A student who replaces `rle03_entry_point_inventory.sh` with their own PowerShell, Python, Go or Rust version producing the same register loses no marks. They gain respect, and reach Lab 05 with a tool they understand completely.

Three conditions only: (1) **achieve the same control outcome** — a different route to the same destination; (2) **record the swap in `CHANGE_LOG.md`** — *what* and *why*, one or two sentences; (3) **own it at validation time** — if your substitute cannot be demonstrated working, that is on you.

Substituting is engineering. Substituting silently is how audits fail.

Research thoroughly before choosing. Read the vendor's own documentation, not a blog summary of it. Check the licence — in industry, licence is an architectural decision, not a footnote. Verify the current stable version on the project's own page and **date the verification**: "*Suricata 8.x, verified 2026-09-06*" beats an unverifiable claim every time. Free, open starting points: <https://www.cisecurity.org/controls> · <https://attack.mitre.org> · <https://www.nist.gov/cyberframework> · <https://docs.opnsense.org> · <https://nmap.org>

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